

Luke Ritter

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Flagstaff, AZ

EDUCATION

Master's of Forestry May 2025

Northern Arizona University, Flagstaff, AZ

Bachelor's of Forestry, Cum Laude May 2024

Northern Arizona University, Flagstaff, AZ

- Forest Health and Restoration certificate

Associates of Fine Arts May 2022

Yavapai Community College, Clarkdale, AZ

TEACHING EXPERIENCE

Co-Instructor, Hat Ranch Field Course (SCI240) May - July; 2024, 2025, 2026

Northern Arizona University

- Collaborated with a team of graduate students and instructors to design interdisciplinary scientific curriculum: forestry, geology, geography, astronomy, wildlife, systems-based, science communication
- Collaborated with a team to allocate a \$1 million grant for purchasing scientific equipment and essential supplies for the development of a field course
- Managed complex social, personal safety, and logistical issues between students
- Worked with PLACES (Place-Based Learning to Advance Connections, Education, and Stewardship) to further teacher understanding of natural sciences and place-based learning.

Instructor, Forest Measurements (FOR211) July - Aug. 2025

Northern Arizona University

Instructor, Human Anatomy/Physiology II (BIO202) Jan. 2025 - May. 2025

Northern Arizona University

Graduate Teaching Assistant, Forest Measurements (FOR211) Aug. 2024 - Dec. 2024

Northern Arizona University

WORK EXPERIENCE (non-teaching)

Consulting Utility Forester

Sept. 2025 - April 2026

Eocene Environmental Group | Arizona Public Service

- Coordinated with customers, stakeholders, and contractors to manage vegetation around utility infrastructure
- Monitored vegetation management for environmental compliance

Field LiDAR Technician

July 2024 - Aug. 2024

Ironwood Forestry, Pumphouse Project, Coconino National Forest

- Used trimbleGPS hardware and handheld LiDAR scanners to collect census-level data
- Used Cloudcompare and QGIS software to analyze and rectify point clouds from scans
- Communicated with LiDAR experts and Forest Service silviculturist

Arizona Wildfire Initiative Student Intern

Feb. 2024 - June 2024

Northern Arizona University

- Engaged in weekly meetings with a focus on research practices and inclusivity in forestry
- Applied funding to graduate research and presentations

Forest Engineering Intern

May 2023 - Aug. 2023

Weyerhaeuser, Vail, WA

- Conducted unit and road layout, water typing, and assessed historic land surveys/records
- Managed data on timber volume and harvest operations
- Supported company operations in compliance with Washington state DNR guidelines

Logging Operations Inspection

Oct. 2022 - March 2023

Ironwood Forestry, Coconino & Kaibab National Forests

- Monitored restoration-focused logging operations according to National Forest guidelines
- Enforced safety and operations standards according to Forest Service guidelines, contracts, and prescriptions

Timber Sale Preparation

May 2022 - Aug. 2024

Ironwood Forestry, Coconino & Kaibab National Forests

- Worked as a crew member to fulfill forest pre-harvest thinning prescriptions

Lead Mechanic

June 2018 - Aug. 2021

Verde Valley Bicycle Company, Cottonwood, AZ

- Coordinated a team of four mechanics and salespeople to repair and sell bicycles and accessories
- Co-planned community outreach and charity events in partnership with Bikes4Kids and Verde Valley Cyclists Coalition

PUBLICATIONS

Ritter, L., Andrew Uhey, D., & Saxena, A. (2025). **Knowledge, attitudes, and practices of forestry professionals towards artificial intelligence (AI)**. *Forest Policy and Economics*, 179, 103626. <https://doi.org/10.1016/j.forpol.2025.103626> Oct. 2025

Related articles:

- **How could AI help (and hurt) forestry?** - The NAU Review, Jill Kimball
- **NAU research team has found uses for artificial intelligence in forestry...** - KJZZ, Fatima Gabir
- **Not only you can prevent wildfires: Northern Arizona Univ. study finds AI could automate forestry tasks** - COAI

Teaching with AI - fact sheet Oct. 2025

Barriers to AI implementation in forestry - fact sheet

Southwest Fire Science Consortium

Embracing the Future: Innovations in Forestry May 2024

The Forestry Source

PRESENTATIONS

Sustainability Dilemmas of AI in Natural Resource Management Oct. 2025

Intro to Sustainable Communities (SUS101), *Northern Arizona University*

Embracing the Future: Innovations in Forestry May 2024

Society of American Foresters, *ForestrySource* commentary article

Technological Innovations in Forestry March 2024

Northern Arizona University, Department of Forestry seminar series

Society of American Foresters Presentation and Q&A March 2024

SAF Southwest Chapter, Flagstaff, AZ

"Hackathon" Subject Matter Expert Sept. 2023

Massachusetts Institute of Technology, Cambridge, MA

CERTIFICATIONS

- Institutional Review Board, Collaborative Institutional Training Initiative (CITI)
- High occupancy vehicle certified, *Northern Arizona University*
- *Safesport* Core training
- AutoCAD drafting
- Class D-M driver's license

PROFICIENCIES

- ATLAS.TI
- ESRI ArcGIS / QGIS
- Cloudcompare
- R markdown (RMD)
- Google product family
- Microsoft product family
- Ascend sales system
- imageJ

Volunteer Experience

Rock Climbing Instructor Oct. 2024 - Ongoing
Flagstaff Climbing, Flagstaff, AZ

- Instructed parents and children ages 4-12 in a rock climbing gym (volunteer)

Volunteer trail building 2018-2020
Cottonwood, AZ

- Worked within state parks to construct and maintain trails using hand tools



Forestry Graduate Student Association, Northern Arizona University Aug. 2024 - May 2025



NAU Cycling Team Aug. 2020 - May 2025



Society of American Foresters Member, Southwest Chapter Aug. 2022 - Ongoing